SmartBrew AI - Group 28 - Lab 2 correction

SmartBrew AI is a personalized coffee-machine system for office customers and maintenance technicians. The full requirements cover profiles, saved recipes, context/mood recommendations, customization, supplies, payments, receipts, diagnostics, privacy and service-failure fallback. The group's custom feature is loyalty redemption. The executable prototype implements ten selected feature groups with persistent local state and a browser interface.

Feedback: positive overall. Main correction: requirements now include Inputs, Outputs, Precondition and Postcondition, following Lecture 2 slide 47. Diagram layout, customer entry, saved descriptions and optional relationships are repaired. Thirteen original requirement/story IDs are retained; added clarifications are documented.

User stories

ID	Name	Story
US-01	Browse Menu and Nutrition	As a Customer, I want to browse categories, prices and nutrition so that I can select a suitable available beverage.
US-02	User Authentication	As a Customer, I want to create an account and log in using credentials or my registered NFC card so that I can access my profile.
US-03	Profile Management	As a Customer, I want to update preferences and save, name and modify custom recipes so that I can reorder my drinks conveniently.
US-04	Beverage Customization	As a Customer, I want to customize milk, strength, temperature, syrup and extra shots so that the drink matches my taste.
US-05	Ingredient Availability	As a Customer, I want unavailable drink selections blocked so that I do not order a drink the machine cannot prepare.
US-06	Personalized Recommendations	As a Customer, I want recommendations using history, time and weather with a local/manual fallback so that I can choose a relevant drink during service outages too.
US-07	Mood Analysis	As a Customer, I want optional mood-based recommendations with immediate image disposal so that suggestions can reflect my mood without retaining my face.
US-08	Natural Language Ordering	As a Customer, I want an optional local LLM to interpret my request so that I can confirm a custom drink using natural language.
US-09	Payment Processing	As a Customer, I want supported payments, honest offline status and a record of the purchased settings so that I can pay reliably and inspect completed purchases.

ID	Name	Story
US-10	Loyalty Point Redemption	As a Customer, I want to earn and redeem loyalty points so that repeat purchases can reduce the price of future drinks.
US-11	Receipt Delivery	As a Customer, I want QR or email receipts so that I can retain proof of purchase.
US-12	Inventory and Waste Management	As a Technician, I want secure access to refills and waste handling, including overflow halts, so that the machine operates safely.
US-13	Error Logging and Diagnostics	As a Technician, I want real-time local error logs and authorized remote diagnostics so that I can investigate faults.

Requirements - lecture template

The six columns follow Lecture 2 slide 47. Functional and non-functional items are labeled separately in Lab 3. For grouped Lab 2 items, labeled conditions apply to the corresponding operation.

ID	Description	Inputs	Outputs	Precondition	Postcondition
REQ-01	Browse Menu and Nutrition The system shall display beverage categories, ingredients, allergens, nutrition, prices and availability.	Category/filter; menu and stock data.	Matching drinks and nutritional details.	Local menu is available.	Customer can inspect drinks; unavailable items cannot be ordered.
REQ-02	User Authentication Create an account: The system shall register a customer account with unique credentials and a preference profile. Authenticate customers: The system shall authenticate credentials or a registered NFC/card identifier and load only the corresponding customer's profile.	Create an account: Email, password, display name. Authenticate customers: Credentials or NFC identifier.	Create an account: Account identifier or validation error. Authenticate customers: Authenticated session/profile or denial.	Create an account: Email is not already registered; required values are valid. Authenticate customers: An account exists; supplied credential is registered.	Create an account: One account/profile is stored; invalid registration makes no change. Authenticate customers: Successful login creates a session; failed login grants no access.
REQ-03	Profile Management Maintain preferences: The system shall let authenticated customers update their default milk, strength and temperature. Manage named recipes: The system shall save, rename, modify and delete personal recipes and restore their settings for one-touch ordering, subject to current stock checks.	Maintain preferences: Customer session; preference values. Manage named recipes: Customer session, recipe name/settings.	Maintain preferences: Saved preference profile. Manage named recipes: Stored recipe or restored order draft.	Maintain preferences: Customer is authenticated; settings are supported. Manage named recipes: Customer is authenticated; recipe settings are valid.	Maintain preferences: Updated defaults persist across logout/restart. Manage named recipes: Only the owner can change a recipe; later orders recheck availability.
REQ-04	Beverage Customization The system shall validate milk, coffee strength, temperature, syrup and extra-shot choices and update the recipe, price and nutrition before confirmation.	Base drink and supported customization choices.	Validated recipe, price, nutrition or error.	A base drink has been selected.	The exact validated settings are visible; invalid options cannot be confirmed.

ID	Description	Inputs	Outputs	Precondition	Postcondition
REQ-05	Ingredient Availability The system shall reject menu selections and confirmed recipes if an ingredient is depleted, below its minimum threshold or insufficient for the requested serving.	Recipe quantities; stock levels/thresholds.	Availability result and missing-ingredient reasons.	Recipe and current counters exist.	Unavailable recipes cannot be purchased; stock is revalidated at checkout.
REQ-06	Personalized Recommendations Rank recommendations: The system shall score available drinks using preferences, purchase history, time and weather, then reorder and highlight the best recommendations. Keep manual ordering available: When network, weather or optional AI services fail, the system shall continue basic local menu/customization subject to ingredient, payment and safety rules.	Rank recommendations: Profile/history, time, weather, available menu. Keep manual ordering available: Unavailable service; local menu/profile/stock.	Rank recommendations: Ranked available drinks and recommendation reasons. Keep manual ordering available: Manual ordering interface and fallback indication.	Rank recommendations: Local menu exists; history/profile may be empty. Keep manual ordering available: Local machine hardware and required supplies are operational.	Rank recommendations: Only available drinks are recommended; missing weather falls back to local context. Keep manual ordering available: External-service loss alone does not block local selection/customization.
REQ-07	Mood Analysis Analyze mood: The system shall use the embedded camera to infer a mood and optionally add it to the recommendation context; unavailable inference shall not prevent manual ordering. Protect facial-image privacy: All camera frames shall be processed only in volatile memory, discarded immediately after inference and never written to disk, logs, databases or external servers.	Analyze mood: Consented transient camera frame. Protect facial-image privacy: Transient camera frame and inference lifecycle.	Analyze mood: Mood label or unavailable result. Protect facial-image privacy: Mood label without image retention/transmission.	Analyze mood: Camera/mood module enabled and customer opts in (design assumption). Protect facial-image privacy: Any camera-based inference runs.	Analyze mood: Recommendations can use the label; no image is retained. Protect facial-image privacy: No facial-image bytes remain in persistent storage or outbound traffic.
REQ-08	Natural Language Ordering When the optional local LLM is enabled, the system shall translate a free-form request into supported recipe parameters and ask for confirmation or clarification.	Request text; local model; supported menu/options.	Validated recipe proposal or clarification question.	Optional module is installed and enabled.	Only customer-confirmed and validated parameters reach ordering.

ID	Description	Inputs	Outputs	Precondition	Postcondition
REQ-09	Payment Processing Process supported payments: The system shall support EMV chip/contactless credit/debit, Apple Pay, Google Pay and prepaid employee cards and distinguish approved, declined and pending outcomes. Record completed transactions: The system shall record each completed purchase once with owner/guest reference, timestamp, original recipe settings, price and payment outcome and show the owner's history. Handle offline payment honestly: The system shall use an approved offline-capable method or securely queue permitted pending payments and reconcile once on reconnect; it shall never present an unapproved online-only wallet payment as successful.	Process supported payments: Confirmed recipe; method; provider response/prepaid balance. Record completed transactions: Approved purchase; recipe snapshot; idempotency key. Handle offline payment honestly: Connectivity; method policy; payment reference.	Process supported payments: Payment result and transaction reference. Record completed transactions: Persistent completed transaction/history entry. Handle offline payment honestly: Offline approval, pending entry or denial; reconciliation result.	Process supported payments: Recipe is available; machine is not halted. Record completed transactions: Payment approved; checkout validations still hold. Handle offline payment honestly: Payment method supports the chosen offline policy.	Process supported payments: Only approved payment completes an order; declines change no stock, points or balance. Record completed transactions: Receipt/history preserve the purchased settings even after a saved recipe is edited. Handle offline payment honestly: Pending authorizations do not dispense; approved prepaid orders and queued diagnostics persist.
REQ-10	Loyalty Point Redemption The system shall let logged-in customers redeem earned points at checkout and award points after a completed paid purchase, with no changes on decline.	Customer balance; requested points; order total.	Discount, paid total and updated points.	Logged-in customer has enough points; redemption does not exceed price.	Spend/earn is committed once with purchase; a rejected purchase leaves points unchanged.
REQ-11	Receipt Delivery After a completed purchase, the system shall offer a QR receipt or email delivery and offer QR as fallback on email failure.	Purchase reference; selected channel; email if needed.	Receipt QR/link or email-delivery result.	A completed transaction exists; recipient choice is valid.	Receipt reflects the immutable transaction; a failed email does not lose the purchase.

ID	Description	Inputs	Outputs	Precondition	Postcondition
REQ-12	Inventory and Waste Management Authenticate maintenance personnel: The system shall restrict administrative actions to technicians authenticated by physical key, registered master NFC card or technician PIN. Register refills: The system shall allow authenticated technicians to register full ingredient/cup refills and reset the selected counter to its configured maximum. Monitor waste and halt safely: The system shall monitor liquid and grounds bins, warn near capacity and halt dispensing on full/overflow status until a technician records emptying.	Authenticate maintenance personnel: Technician PIN/master card/key signal. Register refills: Admin session; selected ingredient; refill confirmation. Monitor waste and halt safely: Waste sensor/counter readings; emptying confirmation.	Authenticate maintenance personnel: Admin session or access denial. Register refills: Updated counter and audit event. Monitor waste and halt safely: Warning/halt status and event log.	Authenticate maintenance personnel: Credential/key is registered for technician role. Register refills: Technician has performed/confirmed the physical refill. Monitor waste and halt safely: Bin capacities and current readings are known.	Authenticate maintenance personnel: Customers cannot refill ingredients, empty bins or read administrative diagnostics. Register refills: Counter equals maximum; unrelated stock is unchanged; availability is recalculated. Monitor waste and halt safely: A full bin blocks ordering; authorized emptying resets that bin and re-evaluates the halt.
REQ-13	Error Logging and Diagnostics The system shall show real-time faults on the admin dashboard and persist diagnostic events for authorized remote diagnostics when connectivity is available.	Fault code, message, severity, timestamp; diagnostic request.	Dashboard event and diagnostic log/export.	Fault reporting is active; diagnostics access is authorized.	Events remain available locally during a remote outage.

Initial product backlog

Importance: proposed 0-100 ordering score, higher first. Estimate: relative story points, not hours. Initial planning values require group review. Description supplements the slide's fields; Notes provides traceability. No sprint commitment is made without team capacity.

ID	Name	Importance	Estimate	Description	How to demo	Notes
PB-02	User Authentication	100	3	Create an account: The system shall register a customer account with unique credentials and a preference profile. Authenticate customers: The system shall authenticate credentials or a registered NFC/card identifier and load only the corresponding customer's profile.	Create an account: Register a new email; retry the same email and verify rejection. Authenticate customers: Use password and DEMO-NFC-001; wrong password is denied.	REQ-02; US-02
PB-05	Ingredient Availability	100	5	The system shall reject menu selections and confirmed recipes if an ingredient is depleted, below its minimum threshold or insufficient for the requested serving.	Refill then exhaust milk; verify milk drink blocked and black coffee still available.	REQ-05; US-05

ID	Name	Importance	Estimate	Description	How to demo	Notes
PB-09	Payment Processing	100	13	Process supported payments: The system shall support EMV chip/contactless credit/debit, Apple Pay, Google Pay and prepaid employee cards and distinguish approved, declined and pending outcomes. Record completed transactions: The system shall record each completed purchase once with owner/guest reference, timestamp, original recipe settings, price and payment outcome and show the owner's history. Handle offline payment honestly: The system shall use an approved offline-capable method or securely queue permitted pending payments and reconcile once on reconnect; it shall never present an unapproved online-only wallet payment as successful.	Process supported payments: Run each method using the simulator; decline one; prepaid insufficient funds is denied. Record completed transactions: Buy an oat recipe, edit it to dairy and verify old history remains oat; retry checkout key and verify one charge. Handle offline payment honestly: Offline prepaid succeeds; wallet stays pending without stock/points changes; reconnect reconciles once.	REQ-09; US-09
PB-12	Inventory and Waste Management	100	13	Authenticate maintenance personnel: The system shall restrict administrative actions to technicians authenticated by physical key, registered master NFC card or technician PIN. Register refills: The system shall allow authenticated technicians to register full ingredient/cup refills and reset the selected counter to its configured maximum. Monitor waste and halt safely: The system shall monitor liquid and grounds bins, warn near capacity and halt dispensing on full/overflow status until a technician records emptying.	Authenticate maintenance personnel: Technician PIN creates admin session; customer token on maintenance endpoint is denied. Register refills: Refill milk and cups; verify exact capacity and audit event. Monitor waste and halt safely: Simulate full grounds, verify checkout blocked, empty grounds and verify ordering resumes.	REQ-12; US-12

ID	Name	Importance	Estimate	Description	How to demo	Notes
PB-01	Browse Menu and Nutrition	95	3	The system shall display beverage categories, ingredients, allergens, nutrition, prices and availability.	Filter Coffee and Chocolate; inspect calories/allergens and availability.	REQ-01; US-01
PB-04	Beverage Customization	95	5	The system shall validate milk, coffee strength, temperature, syrup and extra-shot choices and update the recipe, price and nutrition before confirmation.	Choose Latte/oat/strong/75 C/two syrup shots; verify quote and invalid value rejection.	REQ-04; US-04
PB-03	Profile Management	85	5	Maintain preferences: The system shall let authenticated customers update their default milk, strength and temperature. Manage named recipes: The system shall save, rename, modify and delete personal recipes and restore their settings for one-touch ordering, subject to current stock checks.	Maintain preferences: Save oat milk/strong/75 C, log out/in and verify retained settings. Manage named recipes: Save Morning Oat, rename/edit it and order it again; another user cannot edit it.	REQ-03; US-03
PB-13	Error Logging and Diagnostics	80	5	The system shall show real-time faults on the admin dashboard and persist diagnostic events for authorized remote diagnostics when connectivity is available.	Inject a grinder jam; verify time/severity/local log and authorized JSON export.	REQ-13; US-13
PB-06	Personalized Recommendations	75	8	Rank recommendations: The system shall score available drinks using preferences, purchase history, time and weather, then reorder and highlight the best recommendations. Keep manual ordering available: When network, weather or optional AI services fail, the system shall continue basic local menu/customization subject to ingredient, payment and safety rules.	Rank recommendations: Switch cold/hot and morning/afternoon context; buy a drink and verify history affects ranking. Keep manual ordering available: Disable network/weather; browse/customize and buy using prepaid balance.	REQ-06; US-06

ID	Name	Importance	Estimate	Description	How to demo	Notes
PB-10	Loyalty Point Redemption	70	5	The system shall let logged-in customers redeem earned points at checkout and award points after a completed paid purchase, with no changes on decline.	Redeem 100 points, inspect discount/new balance; repeat with declined payment and verify no deduction.	REQ-10; US-10
PB-11	Receipt Delivery	70	5	After a completed purchase, the system shall offer a QR receipt or email delivery and offer QR as fallback on email failure.	Open a real QR; request email in local outbox; fail email and verify QR fallback.	REQ-11; US-11
PB-07	Mood Analysis	45	8	Analyze mood: The system shall use the embedded camera to infer a mood and optionally add it to the recommendation context; unavailable inference shall not prevent manual ordering. Protect facial-image privacy: All camera frames shall be processed only in volatile memory, discarded immediately after inference and never written to disk, logs, databases or external servers.	Analyze mood: With real module enabled, infer tired mood; deny consent and verify manual ordering still works. Protect facial-image privacy: Inspect filesystem/database and capture network traffic during a real inference run.	REQ-07; US-07
PB-08	Natural Language Ordering	30	13	When the optional local LLM is enabled, the system shall translate a free-form request into supported recipe parameters and ask for confirmation or clarification.	Request sweet but low dairy; inspect parsed proposal; reject unsupported ingredients.	REQ-08; US-08

Diagram and use-case correction

Customer and Technician are primary actors on the left; Payment Processor is secondary on the right. Customer starts Customize Beverage. The thirteen saved use cases all have complete textual descriptions.

Process Natural Language Order extends Customize Beverage. Customize Beverage includes Check Ingredient Availability. Analyze Mood optionally extends Generate Recommendations. Redeem Loyalty extends Process Payment. Dashed extension arrows point toward the base; inclusion points toward the included case.

Technician participates independently in Manage Inventory and View Error Logs; the previous mandatory log-view inclusion was removed. The custom-feature note is anchored to Redeem Loyalty.

Prototype money uses integer ore (100 ore = SEK 1); demo prices/nutrition/ingredient quantities are illustrative and are not production dietary specifications.

Loyalty: earn one point per whole SEK actually paid; one point redeems SEK 0.10, bounded by balance and price. Demo Alex begins with 250 points and SEK 200 prepaid credit; new accounts begin with zero.

Customer orders may be placed as guest; profile/recipe/history and prepaid/loyalty features require customer login. Guest receipts use an unguessable capability token.

Mood opt-in and the proposed 2-second response target are design refinements; the original brief does not prescribe consent UI, a numerical latency target or point-conversion rules.

Sources: supplied system description/guidelines and feedback page 1; Lecture-02-2026 slides 20-23, 30-31, 38, 43 and 47-48. The original submission remains in docs/baseline. Reading the assigned Scrum chapters remains the group's responsibility.

Corrected Astah diagram

